MC124 – Data Structures and Algorithms (3-0-2-4)

Instructor: Dr. Manik Lal Das
Office: 2103, Faculty Block 2,
Email: maniklal_das@daiict.ac.in

Objectives: The course aims to discuss on the fundamental concepts of data structures and algorithms covering linear and non-linear data structures, and complexity analysis of the algorithms to implement some data structures.

Course Contents

- Introduction to Data representation; Linear and non-linear data structures, Iteration, Recursion, Problem solving approaches, Algorithm analysis, running time.
- Arrays; Lists; Stack; Queue; Divide and Conquer approach; Sorting algorithms.
- Binary Tree; Binary Search Tree; Tree traversals; Balanced Tree; Multi-way Search Tree; B-Tree; Heap.
- Graphs; Graph representation; Graph traversals; Depth First Search, Breadth First Search, Spanning Trees; Greedy approach; Shortest path algorithm.
- Dynamic Programming, Matrix multiplications; Backtracking; Knapsack.

Course Outcomes: After completion of this course, the student will be able to:

- Apply suitable data structures in algorithm design and applications;
- Measure performance of data structures and apply skills in algorithm design;
- Solve problems using arrays, lists, stacks, queues, trees, and graphs.

Books:

Data Structures and Algorithms in C++ -- Goodrich, Tamassia, and Mount, (Wiley).
Introduction to Algorithms -- Cormen, Leiserson, Rivest, and Stein, (PHI).

Grading Policy:

In-semester exams – 30%
Lab evaluation – 20%
Quiz, class participation – 10%
End Semester exam – 40%